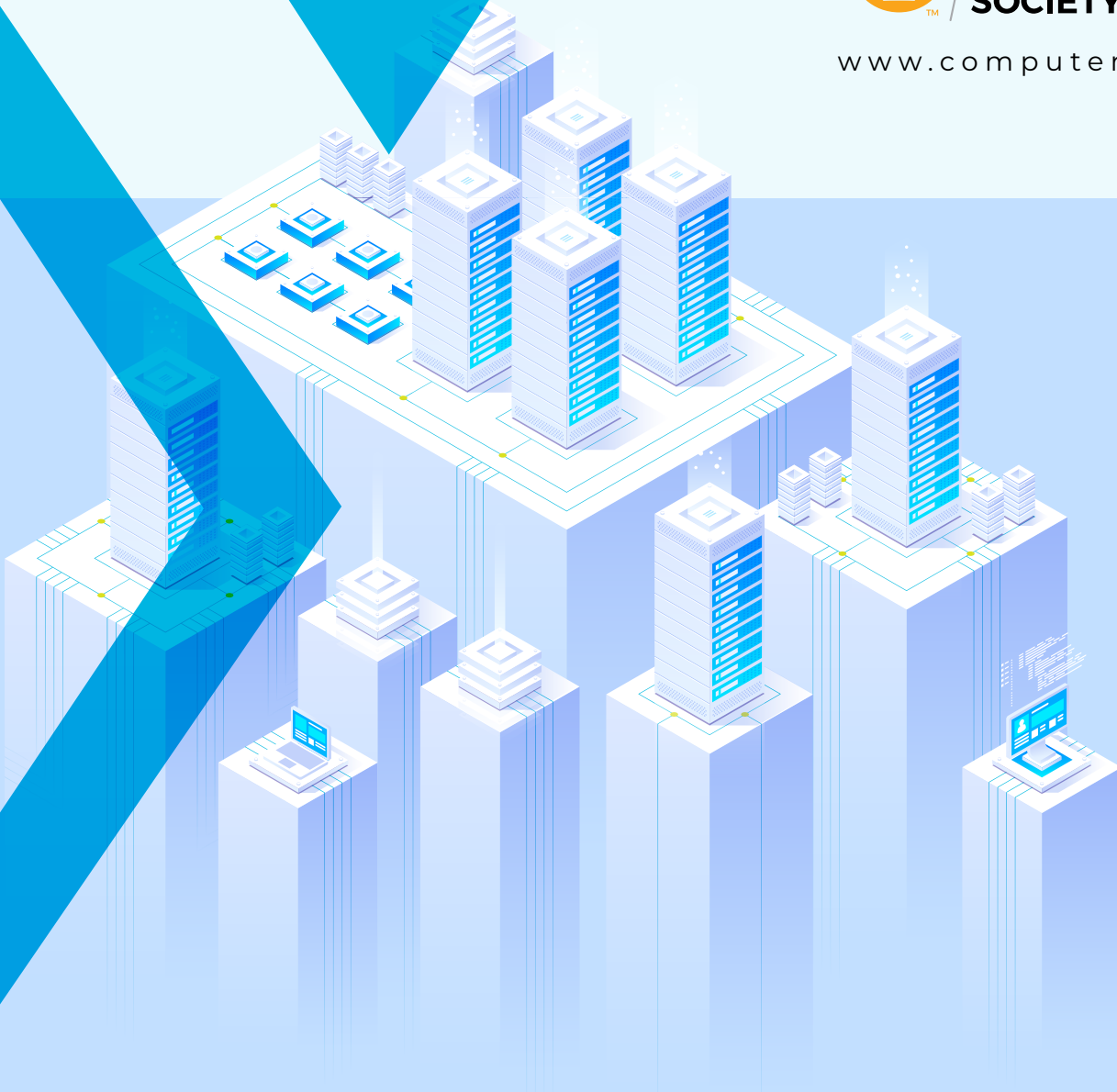




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Cloud Computing: Emerging Opportunities for Engineers

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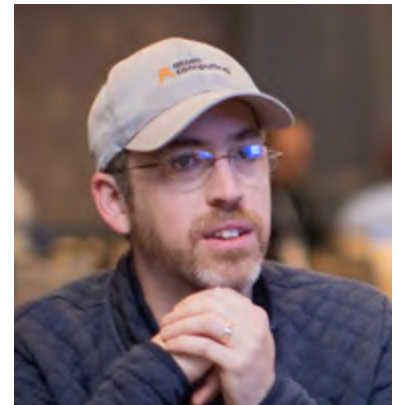
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INTRODUCTION

The significance of cloud computing today is difficult to overstate; the market, which began gaining momentum in 2010, is expected to top US\$1 trillion by 2028. This booming investment and growth in cloud—fed by multinational corporations, the hybrid workforce, and home-based businesses alike—creates enticing job possibilities.



For nobody else is this truer than existing cloud professionals, who bring crucial knowledge and practical experience to a rapidly evolving field. The key to success, according to experts, is two-fold:
be agile and be prepared.



San Murugesan, adjunct professor at Western Sydney University's School of Computing, director of BRITE Professional Services, and former editor-in-chief of IEEE Computer Society's *IT Professional* magazine, says that in cloud computing today, relevant technical and

operational skills are in high demand across sectors.

"Opportunities abound at all levels, ranging from entry-level to executive positions, and [in] several functions," says Murugesan. "To capture those opportunities, however, early career, as well as

Dr. San Murugesan / Director—BRITE Professional Services



Dr. Murugesan's expertise ranges smart systems, Cloud computing, digital transformations, emerging computing paradigms, and more. He's edited, co-edited, and authored more than 44 books and reports. As a Distinguished Lecturer, some of his presentations include Artificial Intelligence: Applications and Addressing Concerns; and, The Rise of Autonomous Systems: Advances, Applications, and Opportunities.

Kailash Selvaraj / Joint Director—Centre for Development of Advanced Computing (C-DAC)

Kailash holds 16+ years of experience in cloud, grid and high performance computing at C-DAC, a scientific society under the Indian Government's Ministry of Electronics and Information Technology. He's involved in the design of free, open source cloud solutions, and has worked as a cloud architect and consultant for government cloud implementations and a corporate trainer.



established professionals, must keep continually updated with ongoing technological developments and new cloud service offerings—and learn from their own and others' cloud success and failure stories.”

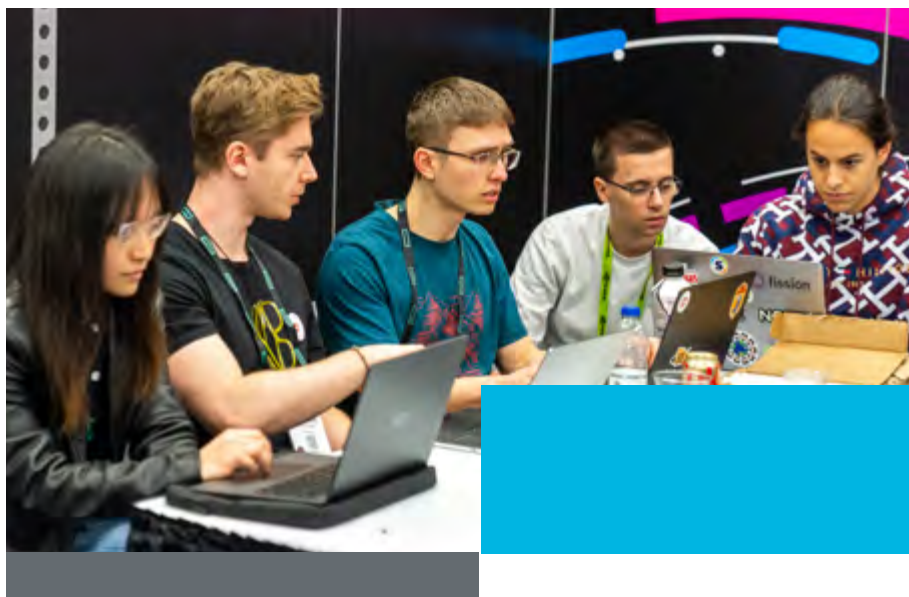
In short, Murugesan says, ambitious cloud professionals “need to stay cloud-fit always and future ready.”

Kailash Selvaraj agrees, noting the sharp contrast in attitudes toward cloud services from the early days of the “cloud journey” to the views of organizations today.

“During the initial period, various organizations were not easily approachable, and it was highly difficult to convince organizations to adopt cloud technology—specifically with free and open source software,” he says, adding that

the scenario changed gradually but clearly over time. Today, almost all organizations are seeking out public or private cloud deployments to serve their business needs.

As a cloud engineer, or just entering the field as an early career professional, it's best to capitalize on the resulting—and plentiful—job opportunities that rest in maximizing your on-the-job experience, training, and networking to ensure you will be prepared for and aware of the next steps in your cloud journey.



Cloud's Growth: Challenges and Opportunities

Although the technological developments in cloud computing have progressed along generally expected lines, what was less expected was how rapidly cloud services took off. Indeed, what began as a technology offering by what was then considered a non-IT company—Amazon—quickly exploded, diversified, and was adopted across the world “by all major players—industry, business, education, and healthcare sectors,” says Murugesan. This rapid growth was fueled by the benefits that cloud services offer, including reduced costs and increased scalability, as well as greater innovation possibilities and access to users.



“Cloud is transforming every sector of our society and the IT industry,” says Murugesan.

“Corporations are eagerly investing in promising cloud computing technologies and services, not only in developed economies but also increasingly in emerging economies—including India, China, Singapore, Taiwan, the Philippines, and South Africa—to address a region’s specific needs.”

While this explosive growth in cloud services has delivered promised benefits

to organizations across the world, it has also created challenges for businesses and IT professionals.

Challenges and Opportunities

To take advantage of cloud career opportunities, cloud engineers must be aware of and **capable of addressing various cloud challenges**, including:

- integrating on-premise and cloud infrastructure and applications,
- deploying applications on multiple clouds,

- securing data and applications, and
- meeting regulatory and service requirements when deploying new and existing applications.

Murugesan says these challenges will soon include a plethora of others. "The rise of 5G communication, high-speed Wi-Fi, and autonomous vehicles such as driverless cars and trucks will drive the cloud to newer heights and will open up new opportunities to all stakeholders," he says.

"Organizations that better understand and harness cloud



Cloud is transforming every sector of our society and the IT industry.



computing's potential, and strengths—while recognizing and addressing limitations and adopting a holistic cloud strategy—stand to gain in the new era of IT.”

Emerging areas in cloud computing, according to Selvaraj, include edge computing and targeted cloud platforms and services for edge

devices such as the Internet of things (IoT). He also highlights **key areas such as hybrid and multi-cloud services integration, as well as zero trust**, which all major providers are now supporting. Security challenges include providing improved data loss prevention and continuity of service in the event of a failure or hack.

Emerging areas in cloud include edge computing and targeted cloud platforms and services for edge devices.



Cloud Career Opportunities



As cloud expands its reach and scope and new services emerge, cloud professionals will see their already stellar opportunities expand. According to Robert Half, head of the Robert Half global HR consulting firm, capitalizing on cloud opportunities requires experience with and certification in one or more cloud platforms (AWS, Azure, or Google Cloud), as well as **experience with APIs and web services and software configuration management**. Among the SCM tools that Half recommends learning are Puppet, Ansible, and Chef. But perhaps most importantly, Half says, cloud professionals must be problem solvers, particularly when it comes to working with organizations who are only

beginning their cloud migration.

As the cloud field continues its expansion, being able to **analyze known and emerging problems**—and find solutions—will be essential to all existing cloud computing roles, as well as to those roles that will emerge in the years to come.

Existing Roles

Whether you are new to the cloud workforce or have a burgeoning portfolio of projects and experiences, identifying your ideal future position and the requirements it entails can help you begin mapping your path to get there. Following are four established, high-level cloud computing positions.

Cloud Architect

In addition to designing, managing, and maintaining an organization's cloud environment, solutions, and services, cloud architects work with various IT and business stakeholders to understand their needs and meet their goals. Along with a solid grasp of IT resources and programming languages—such as Python, Ruby, and Java—cloud infrastructure architects must know how to design and deploy cloud infrastructure, as well as cloud solutions and services, and be familiar with or certified in one or more cloud environments. Because they interact with a variety of stakeholders, all cloud

architects require strong soft skills, including excellent communication skills and leadership abilities. Related professional certifications include AWS Certified Solutions Architect, Azure Solutions Architect Expert, Google Certified Professional Cloud Architect, and Professional Cloud Architect. In the United States, the salary for a cloud architect ranges from an average of US \$150,000 to upwards of US \$190,000 a year.



Cloud Network Engineer

Typical responsibilities for a cloud network engineer include designing, testing, analyzing, and implementing cloud solutions; ensuring that these solutions meet the organization's performance and scalability needs; and, implementing security measures and monitoring network usage. This position often includes extensive interaction and collaboration with network engineering and security operations teams. In addition to a solid grasp of IT resources, Linux, networking tools, and programming languages,

cloud network engineers need extensive experience in information security. The related professional certifications for this position are those from the major cloud providers and VMware's Network Virtualization. In the United States, the average salary for this position is US \$120,000 to upwards of US \$160,000.



Cloud Consultant

A cloud consultant helps organizations understand their cloud options and how various services align with business operations and needs. Cloud consultants must understand cloud technology, implementation, and security issues; be able to describe issues to organizational leaders in lay terms clearly; and, map business needs to specific cloud options and services.

Cloud consultants also collaborate with cloud engineers and administrators, as well as IT teams, to create and integrate cloud applications and to configure existing networks to work with cloud services. Cloud consultants might work in-house as part of a large organization's IT team, or work with various organizations as

contracted consultants, either independently or as part of a consultancy or research firm. Among the top certifications for consultants are Salesforce's Cloud Consultant Certification, as well as solutions architect certifications from AWS and Azure. In the United States, the average salary for this position is US \$131,000 to upwards of US \$170,000.



Cloud Security Analyst

Analysts design and implement cloud security architectures, advise on best practices, review and strengthen existing security measures, and analyze security measures for compliance with cloud standards. Requirements include experience with identity and access management; security issue detection; protection challenges and measures for

cloud networks, applications, and data; and, compliance with cloud security standards and control frameworks. Google, Azure, and AWS offer cloud security certifications; others include Certified Cloud Security Professional, CompTIA Cloud+, and the Cloud Security Alliance's Certificate of Cloud Security Knowledge. In the United States, the average salary for this position is US \$112,000 to upwards of US \$137,000.



Emerging Opportunities

Cloud's rapid growth is leading to new needs and new opportunities. In addition to the expanding opportunities in **critical cloud areas such as edge computing, zero trust, and multicloud services integration**, Selvaraj says emerging roles such as cloud auditor and cloud service brokers will mature and grow in importance over time.

Cloud auditors are typically outside analysts who audit cloud systems for compliance with standards and ensure that subscribing organizations have secure, reliable network access and reasonable costs. Given the exploding data storage market, such assurances are crucial and will only grow in importance over time.

Cloud service brokers act as intermediaries between cloud providers and consumers to simplify service integration and options. Brokers ensure interoperability between cloud services, reduce dependency on individual providers,

and can also offer cost optimizations by purchasing in bulk.

As organizations continue to deepen their investments in and reliance on cloud providers and services, Murugesan predicts new positions will gain prominence, including two senior-level roles that he expects will be needed in the future:

Chief cloud system officers (CCSOs) will be responsible for a company's cloud presence and infrastructure. Murugesan adds that CCSOs will develop and implement cloud strategy; update and enforce company cloud policies; establish security protocols and policies; ensure uptime and business continuity; and assess the need for updates and evolution. This position will also be responsible for assessing and managing potential risks and overseeing compliance with regulatory requirements.

Cloud security and availability officers (CSAOs) will be responsible for end-to-end security, says Murugesan, as well as data storage and continuity of business in the face of cloud failures.

Contributing to the Profession



Although cloud adoption is everywhere, the professional world surrounding and supporting cloud computing remains relatively contained. It is nurtured and sustained through education, knowledge sharing, and networking. In addition to technical skills and experience, **who you learn from, who you know, who you help, and who helps you remain key elements** of a successful career.

Building and contributing to a network of colleagues and mentors begins when you engage with the world in and beyond your workplace. The following are suggestions and options to consider.



On the Job: Always Be Learning

As in any IT job, the key to making the most of your work experience is taking advantage of every opportunity to learn, teach, challenge yourself, and grow—not only as a professional, but as a person. Such opportunities might take the form of working overtime to shadow a colleague, so you can learn a new tool or skill, or taking a public speaking class to up your presentation game. Knowing your strengths and sharing them, understanding your weaknesses, and finding ways to shore them up are great places to start.

Mentorships can also help you improve. Finding a mentor on the job is ideal. Still, you can **establish an informal mentorship** by identifying someone in your targeted position

and studying how they got there, reading their publications, following them on social media platforms, and figuring out what you can learn from their journey.

At a minimum, formal education requirements for mid-level cloud computing positions include a bachelor's degree in computer science, network engineering, information systems, or other IT-related areas. Certification in and experience with at least one of the top three platforms (AWS, Azure, or Google Cloud) is also typically expected. As with most dynamic professions, in cloud computing, **learning gained through real-world experiences** is important and highly valued.

In the World: Conferences and Professional Societies

Attending conferences and participating in professional societies is a great way to learn and grow your professional networks. The following are a few upcoming conferences to consider:

- International Conference on Cloud Computing: Services and Architecture (CLOUD)

- International Conference on Cloud Computing and Big Data Analytics (ICCCBDA)
- International Conference on Cloud Computing and Computer Network (CCCN)
- IEEE International Conference on Cloud Computing (IEEE Cloud)

Joining a professional society has obvious benefits, such as giving you access to online resources and industry news; it also adds to your resume and shows future employers that you're committed

to the field. Such a commitment can be focused solely on your education, or it can extend to supporting others in the community. Professional organizations offer many volunteer opportunities, ranging from peer reviewing for journals to helping out at a sponsored



conference. Such activities offer chances to meet peers and more established professionals in the field.

The following are a few professional organizations to consider:

- [IEEE Computer Society](#)
- [IEEE Computer Society Technical Community on Cloud Computing](#)

- [Cloud Security Alliance](#)
- [Cybersecurity Special Technical Community](#)

Other membership options include a myriad of online groups, ranging from social media-based cloud computing groups on LinkedIn to the community groups associated with the [World Wide Web Consortium \(W3C\)](#), the [Open Group](#), and [techUK](#).



Moving Forward

As expected, the projected short-term path ahead for cloud computing will be littered with challenges. Looking at the near future, Selvaraj says his biggest concern—across infrastructure, application, security, data, and management perspectives—is **ensuring compliance with cloud standards**. From the cloud solution provider's perspective, he says that service interoperability and portability are the chief concerns, while cloud administrators will face issues with instrumentation and control.

The top cloud concern for Murugesan is security. As the cloud landscape widens, cybercriminals will adopt increasingly sophisticated tools and launch novel attacks, creating a perpetual cloud security challenge. "Cloud is—and will continue to be—a fertile and most attractive domain for cyberhackers and criminals," he says. "Hence, cloud vendors, adopters, and users need to be more vigilant than before."



“As businesses recognize the value and vast potential of the cloud, the legacy system will expire, and cloud adoption will not only continue and will become more widespread than now.”

Other challenges that Murugesan sees include performance and reliability guarantees, enhanced access control, and facilitating easier integration of data and applications distributed among multiple clouds from different vendors.

As noted earlier, challenges associated with a burgeoning and ever-evolving field create opportunities for ambitious early-career professionals. Assuming that you continue to hone your skills and engage with the cloud community, there are few imaginable limits to what you can achieve. “The outlook for cloud and cloud computing professionals is bright,” says Murugesan. “As businesses recognize the value and vast potential of the cloud, the legacy system will expire, and cloud adoption will not only continue and will become more widespread than now.”

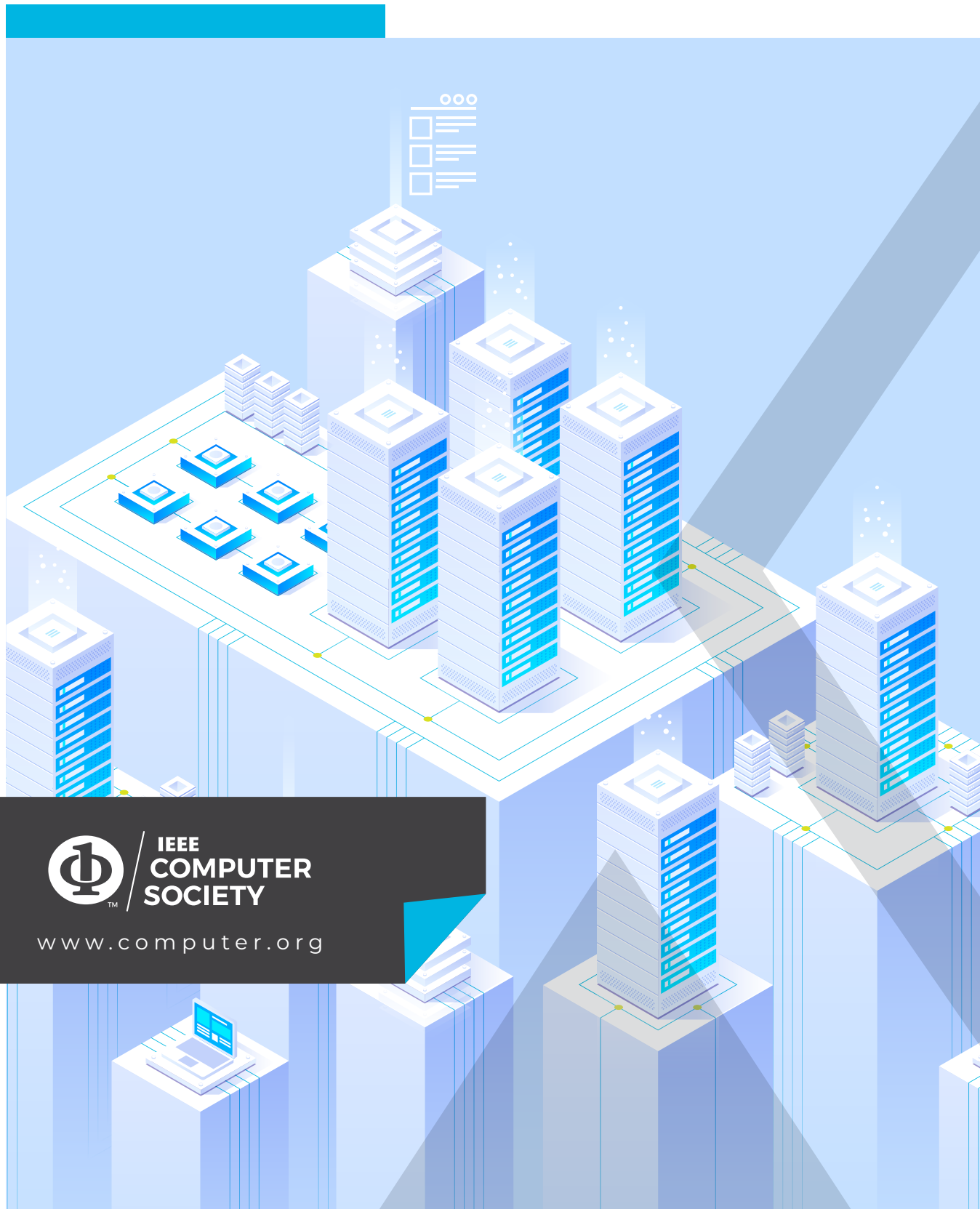


LOOKING AHEAD

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